Physics GCE A Level 2020 - Paper 2

Duration: 2h30 | Cameroon GCE Board

Cameroon General Certificate of Education Board - General Certificate of Education Examination. Physics 2 (0780), ADVANCED LEVEL, June 2020. Time allowed: two and a half hours.

Answer ALL questions. Section I is designed to be answered in 1 hour, Section II in 30 minutes and Section III in 1 hour. You are advised to divide your time accordingly. You are reminded of the necessity for good English and orderly presentation in your answers. In calculations you must show all the steps in your working, giving your answer at each stage. Calculators and formulae booklet are allowed.

Structure of the paper. SECTION I (One hour, answer all questions): questions 1 to 5 are compulsory; in question 6 answer EITHER 6 (a), (b) and (c) OR 6 (d), (e) and (f). SECTION II (30 minutes): DATA ANALYSIS, question 7. SECTION III (1 hour): OPTIONS - answer any two questions from the four options (Option 1: Energy resources and environmental physics, question 8; Option 2: Communication, question 9; Option 3: Electronics, question 10; Option 4: Medical physics, question 11).

No data sheet is printed in the paper: whenever a constant is needed, the model answer states the value it uses.

Exercise 1 (6 marks)

The power, P, obtainable from a wind turbine may be given by $P = KL^2\rho v^3$, where L = length of the blades, ρ = air density and v = wind speed. The equation is homogeneous.

a. What is a homogenous equation?

b. Determine the base units of the constant K.

c. State two shortcomings of using homogeneity to check the correctness of a physical equation.

Exercise 2 (7 marks)

In *figure 1*, a charged metal sphere (A) is hung from an insulating string. Another charged sphere (B) on an insulating rod is then placed close to A as shown.

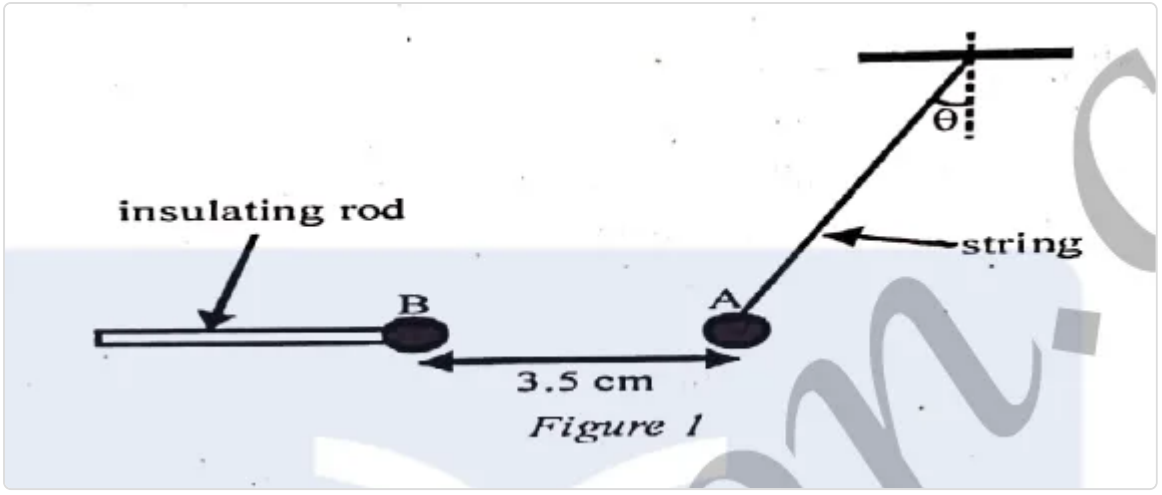


Figure 1

The charge on sphere A is $+5.0\ \text{nC}$ while that on sphere B is $-4.0\ \text{nC}$.

a. Draw the two spheres and show the electric field pattern around them.

b. Determine the magnitude of the electric force between the two spheres.

c. What is the value of the angle θ if sphere A has a mass of $4.5 \times 10^{-5}\ \text{kg}$?

Exercise 3 (7 marks)

Figure 2 shows part of a system used to eject and accelerate electrons.

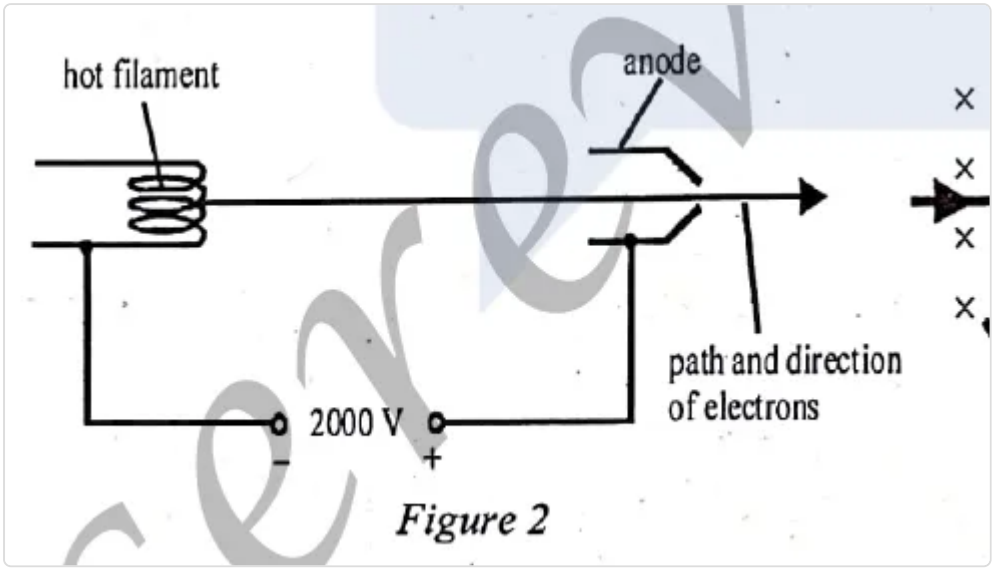


Figure 2

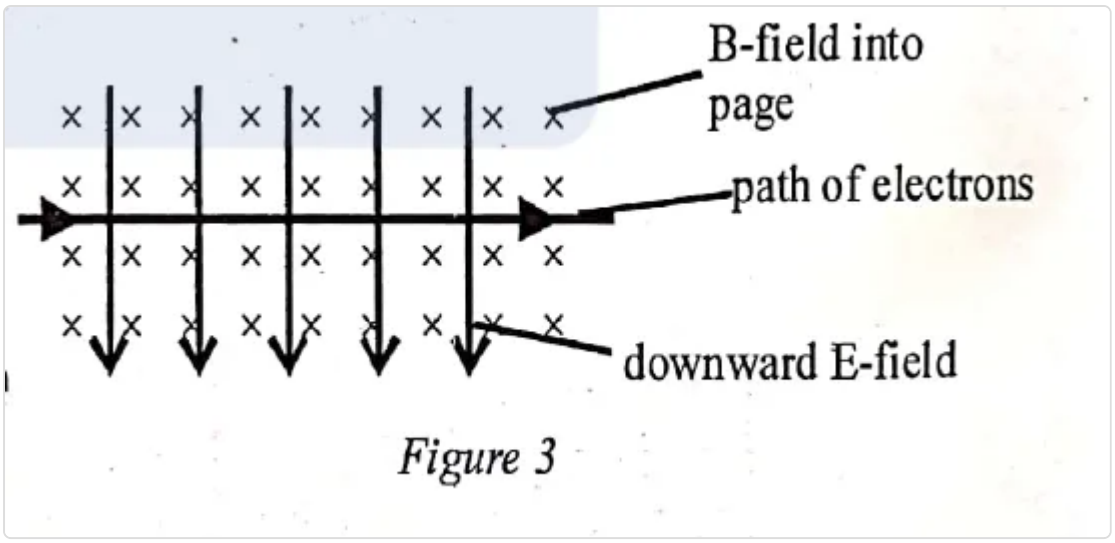


Figure 3

a. State the phenomenon of emission of electrons from a sufficiently heated metal surface.

b. Determine the speed with which the electrons leave the anode.

c. Upon leaving the anode, the electrons enter into a region of cross magnetic and electric fields as shown in *figure 3*. If the intensity of the electric field is $7.5 \times 10^4\ \mathrm{V\,m^{-1}}$, calculate the flux density of the magnetic field if the electrons pass through the cross fields undeflected as shown in figure 3.

Exercise 4 (5 marks)

Figure 4 shows the fringe pattern obtained on a screen from Young's double-slit experiment to measure the wavelength of a **monochromatic light** (underlined in the paper) from a source.

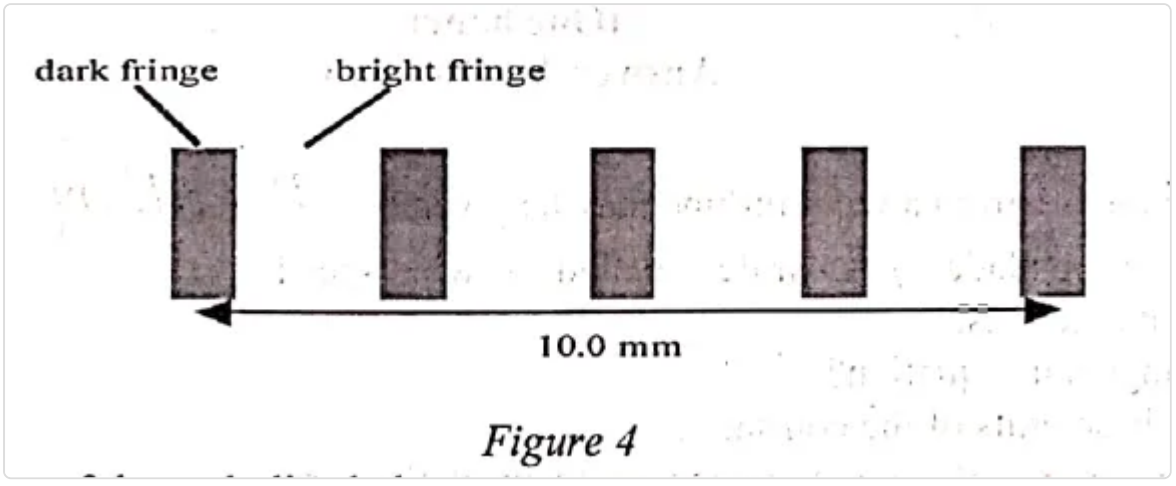


Figure 4

a. Explain the meaning of the underlined phrase.

b. Determine the fringe separation.

c. If the distance from the double slits to the screen is $2.5\ \mathrm{m}$ while the separation of the double slits is $0.50\ \mathrm{mm}$, determine the wavelength of the light from the source.

Exercise 5 (5 marks)

Strontium-89 has a **half-life of 84 days** (underlined in the paper).

a. Explain the meaning of the underlined phrase.

b. A laboratory prepares a strontium-89 source. 21 days after preparation, its activity is measured to be 7.4×10^6 Bq. What is the activity of the source at the time of preparation?

Exercise 6 (20 marks)

Answer EITHER 6 (a), (b) and (c) OR 6 (d), (e) and (f).

a. (i) State the first law of thermodynamics. *[2 marks]*

2.00 kg of water has a volume of 2.00×10^{-3} m^3 when in the liquid state at $100\ ^\circ$C. When the water is completely changed from liquid to vapour at $100\ ^\circ$C, under a constant atmospheric pressure of 1.01×10^5 Pa, the volume increases to 3.38 m^3.

(ii) How much work is done against the atmosphere as the water changes into vapour?

(iii) What is the change in the internal energy of the water during vaporization?

b. The graphs in *figure 5* show how tensile stress varies with tensile strain for two metallic wires, copper and steel. Study the two graphs and answer the questions that follow. The stress is applied until the material breaks, in each case.

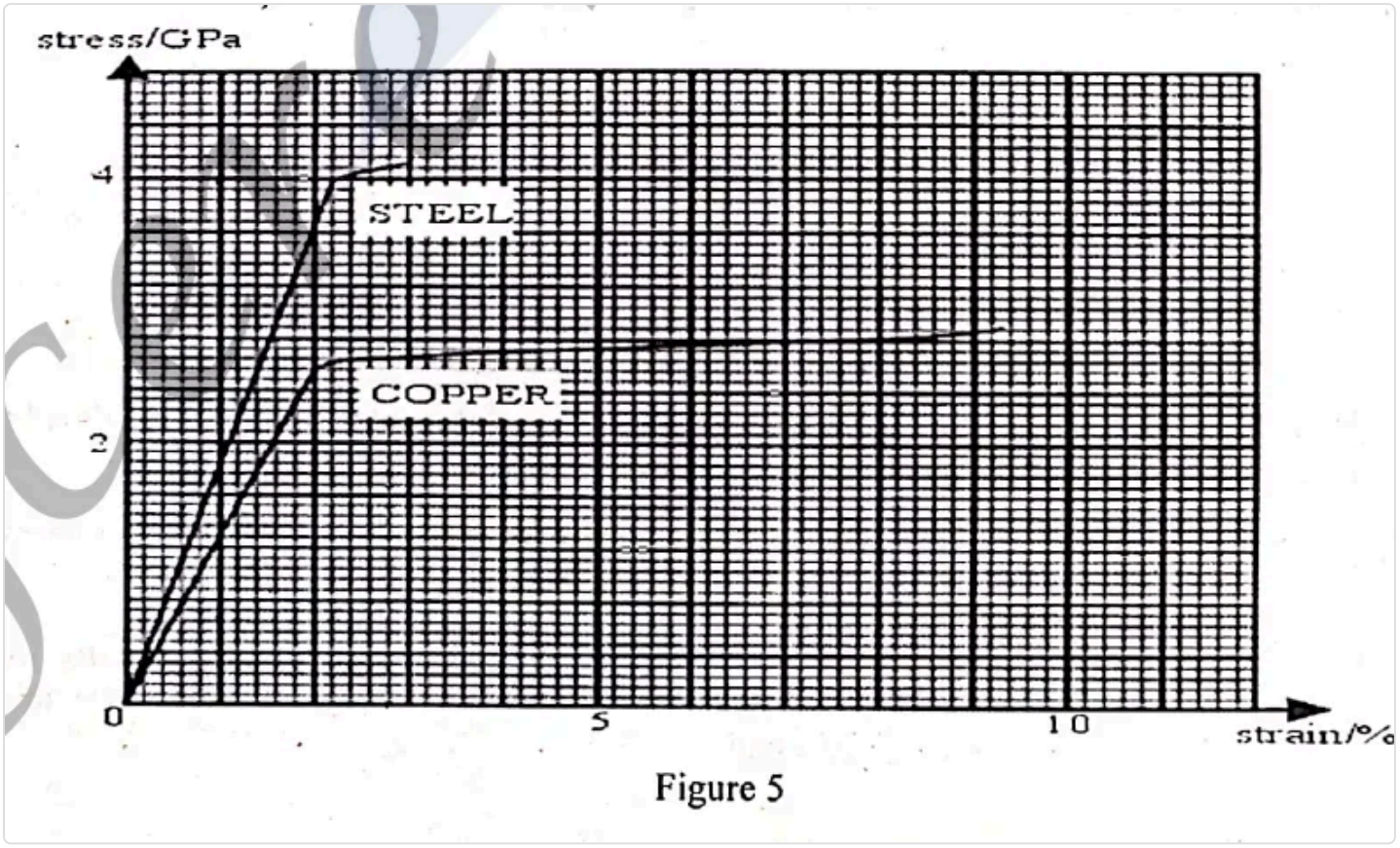


Figure 5

(i) Write down the breaking tensile stress of each of the materials.

(ii) Which of the materials is more ductile? Explain your answer.

(iii) Determine the Young modulus of each of the materials.

c. (i) State any two assumptions of the kinetic theory of matter used to derive the equation $P = \frac{1}{3}\rho\overline{c^2}$.

(ii) What is the name given to $\overline{c^2}$?

d. (i) What do you understand by the *electromotive force of a cell*?

In an experiment to determine the internal resistance of a battery, the current through a battery was made to vary with the voltage across a variable resistor as shown in *figure 7*.

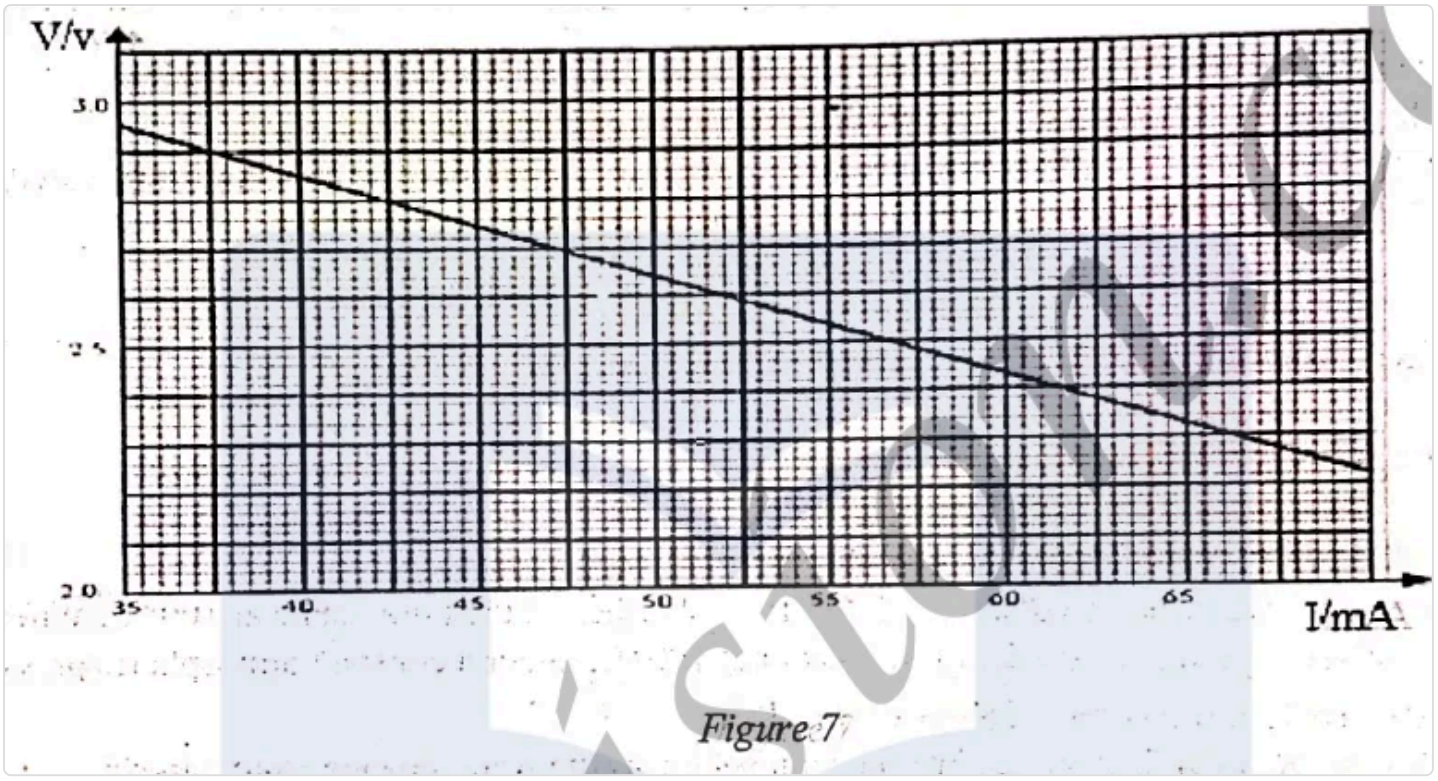


Figure 7

Use the graph in figure 7 to determine:

(ii) the internal resistance of the battery;

(iii) the e.m.f. of the battery.

e. The three lowest energy levels of an atom are shown in *figure 8* below.

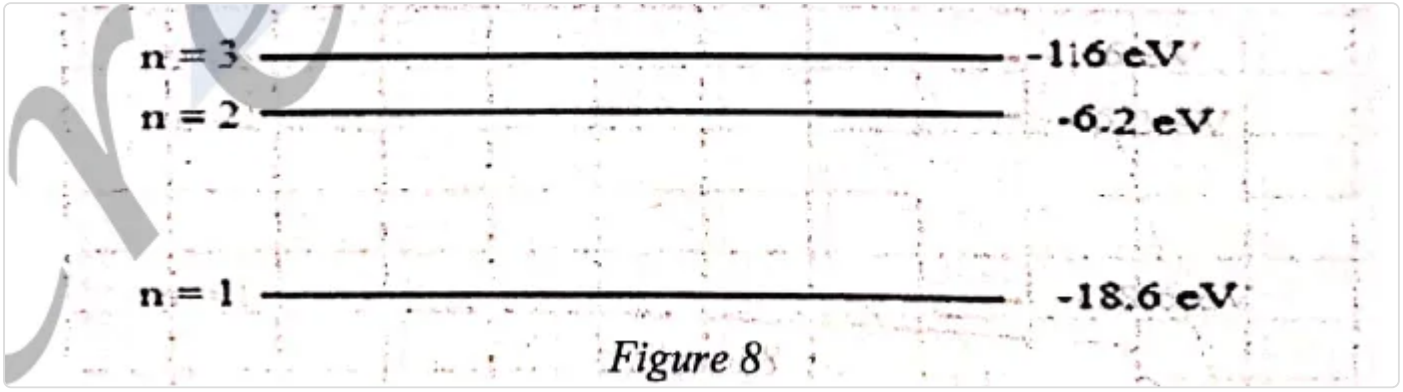


Figure 8

(i) Determine the minimum energy in joules required to eject an electron initially in the lowest energy level from the atom.

(ii) Assuming that the energy level, n, is associated with energy $\frac{k}{n^2}$, determine the energy of the level $n = 4$ in electron-volts.

(iii) Determine the wavelength of the electromagnetic radiation emitted when an electron makes a quantum jump from the level $n = 3$ to the level $n = 2$. Name the region of the electromagnetic spectrum in which this radiation is found.

f. *Figure 9* shows a $200\ \mu\text{H}$ inductor, a $10\ \Omega$ resistor and a variable capacitor connected in series with a $0.10\ \text{V}$ (r.m.s. value), $1.0\ \text{MHz}$ supply. The inductor has an inductance of $200\ \mu\text{H}$.

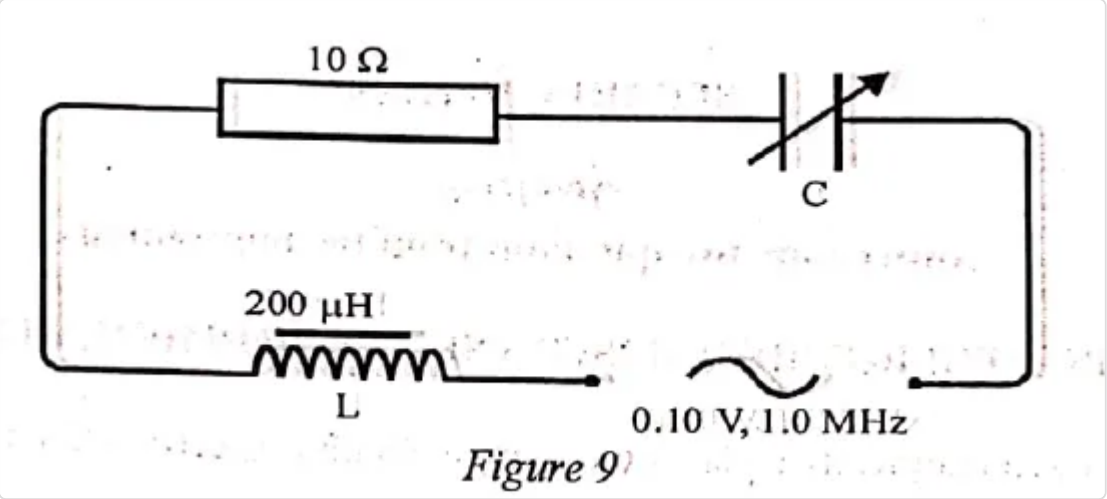


Figure 9

At resonance, determine:

(i) the capacitance of the variable capacitor;

(ii) the current flowing in the circuit;

(iii) the voltage across the inductor;

(iv) the voltage across the capacitor.

SECTION II (30 minutes) - DATA ANALYSIS

In an experiment to investigate how the current, I, in a circuit comprising a capacitor and a resistor in series varies with time, t, a student obtained the following results.

Time, t / s	**I / μA**
10	46.7
40	38.5
70	32.2
100	26.8
130	22.3
160	18.5
190	15.5
220	12.9
250	10.8

Time, t / s	I / μA
280	9.1
310	7.7

The current in the circuit is related to the time by the equation $I = I_0\, e^{-t/\beta}$, where I_0 and β are constants.

a. Plot a suitable graph that can be used to determine the values of I_0 and β. *[10 marks]*

b. Use your graph to determine the values of I_0 and β. *[8 marks]*

c. What is the physical significance of β? *[2 marks]*

Exercise 8 (15 marks)

OPTION 1: ENERGY RESOURCES AND ENVIRONMENTAL PHYSICS

a. *Figure 10* shows a photovoltaic plate used to light a bulb in a certain household.

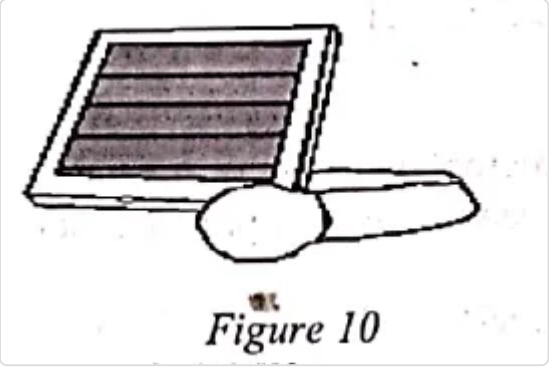


Figure 10

The solar panel has a power output of $50.0\ \mathrm{W}$.

(i) If the efficiency of the panel is $30\ \%$, determine the power input into the photovoltaic plate.

(ii) Above the earth's atmosphere, the solar constant is $1.4\ \mathrm{kW\,m^{-2}}$. However, only $40\ \%$ of this power per unit area reaches the earth. Estimate the surface area of the panel in figure 10.

(iii) Explain why solar energy is referred to as a renewable source of energy. *[6 marks]*

b. (i) What is the ozone layer?

(ii) Name a substance which is responsible for the depletion of the ozone layer and explain how this substance contributes to the depletion of the ozone layer.

(iii) State two effects of the depletion of the ozone layer on life on earth. *[6 marks]*

c. (i) Give two reasons why Cameroon relies more on hydroelectricity power plants for the supply of electricity instead of fusil fuels.

(ii) Give one example of a green source of energy. *[3 marks]*

Exercise 9 (15 marks)

OPTION 2: COMMUNICATION

a. *Figure 11* shows a light signal entering into an optical fibre.

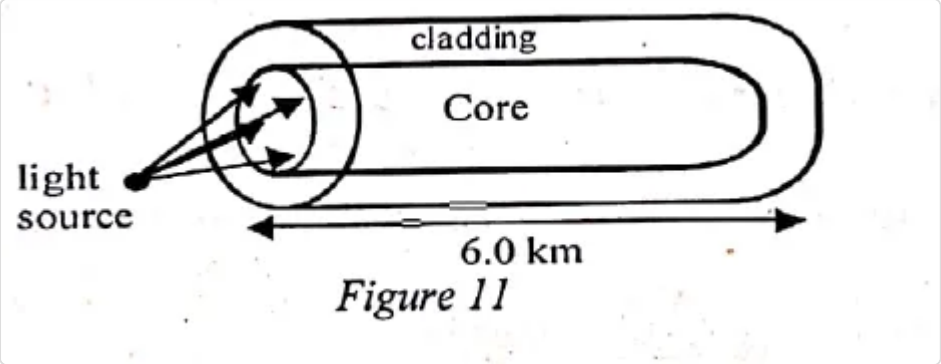


Figure 11

A pulse of light is emitted by switching on the source for $1.0\ \mu s$. The light is emitted in many directions as shown. The critical angle at the core/cladding boundary is 59° and the length of the fibre is $6.0\ km$. The refractive index of the core is 1.50.

(i) State two advantages that optical fibres have over copper cables in telecommunication.

(ii) Some light from the source will travel straight down the fibre for its entire length. Determine how long it takes to travel the entire length of the fibre.

(iii) What is the refractive index of the cladding? *[6 marks]*

b. *Figure 12* shows the block diagram of a simple radio transmitter.

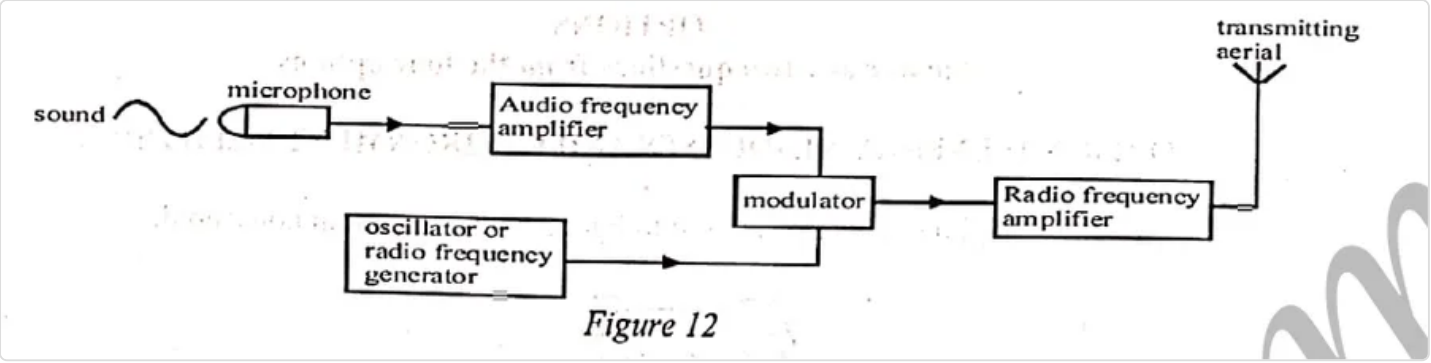


Figure 12

(i) State the function of the microphone and the modulator.

(ii) Briefly explain the difference between amplitude modulation (A.M.) and frequency modulation (F.M.).

(iii) State one advantage that each of F.M. and A.M. has over the other. *[6 marks]*

c. State three advantages of digital transmission over analogue transmission. *[3 marks]*

Exercise 10 (15 marks)

OPTION 3: ELECTRONICS

a. (i) Explain why the electrical conductivity of a semiconductor increases as temperature increases?

(ii) Use the band theory to distinguish between the electrical properties of wood, silicon and copper. *[5 marks]*

b. *Figure 13* shows a photodiode in circuit.

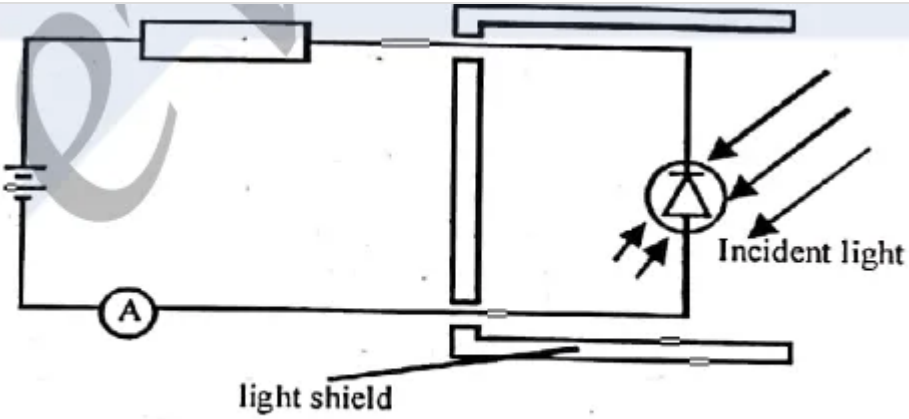


Figure 13

(i) Explain the meaning of dark current of a photodiode. What is the order of magnitude of this current?

(ii) What is the effect of increasing the light intensity on the diode? Explain. *[4 marks]*

c. *Figure 14* shows an n-p-n transistor in a circuit that can be used in voltage amplification.

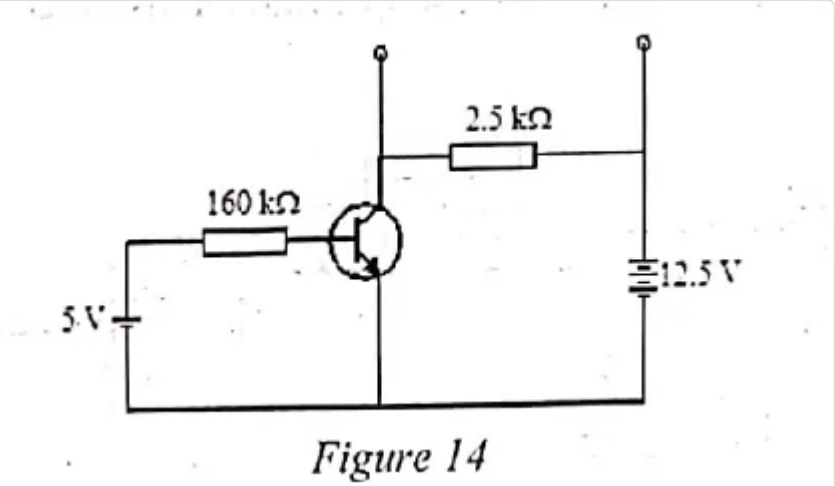


Figure 14

The base-emitter voltage, $V_{BE} = 0.7$ V and current gain (h_{FE}) $= 100$.

(i) Determine the base current and the collector current.

(ii) What is the operating point of the above transistor? *[6 marks]*

Exercise 11 (15 marks)

OPTION 4: MEDICAL PHYSICS

a. (i) Why is it necessary to use a gel where the ultrasonic probe is placed?

(ii) State, with respect to medical diagnosis, one advantage that

- X-rays imaging has over ultrasound imaging

- ultrasound imaging has over X-ray *[5 marks]*

b. (i) Describe the changes that occur to a human eye when the eye changes from focusing a distant object to focusing a near one, both objects being [...] viewed under bright light.

(ii) A patient's eye is said to be astigmatic. State the effect of astigmatism on the image produced by the defective eye, its cause and the shape of the lens used for correction.

A patient's near point is located $210\ \text{cm}$ from his eyes.

(iii) What type of eye defect is the patient suffering from?

(iv) Use diagrams to explain the type of lens needed to correct this defect so that the patient should be able to read a book placed $25\ \text{cm}$ from his eyes. *[7 marks]*

c. *Figure 15* shows the electrocardiogram waveform obtained at the surface of the skin of a patient.

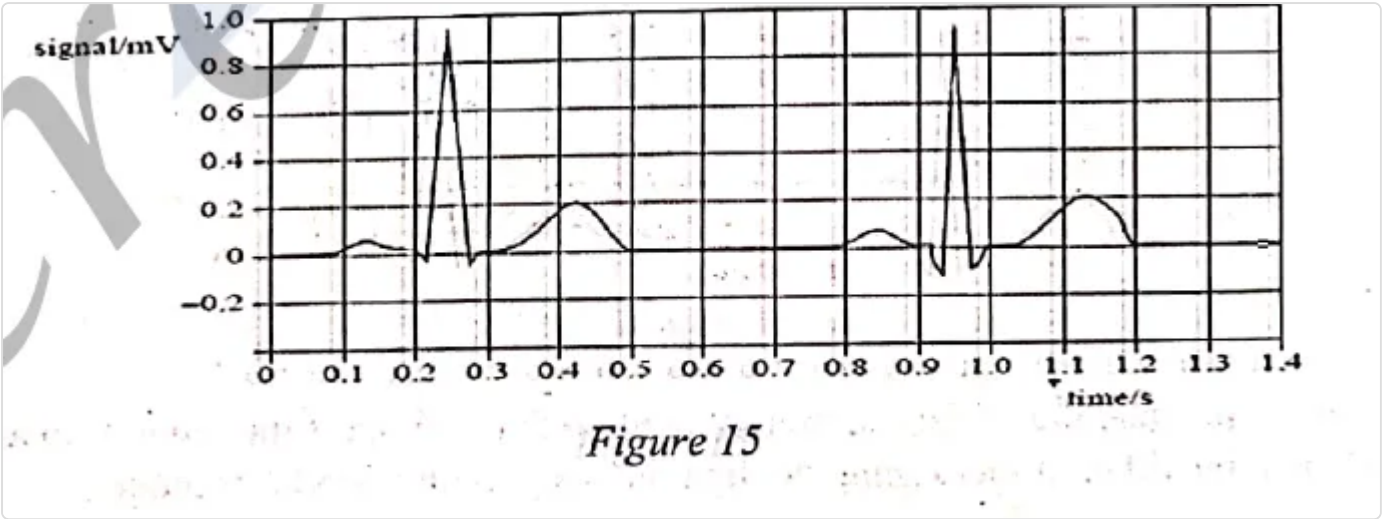


Figure 15

From the ECG waveform, determine the pulse rate per minute. *[3 marks]*

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